Empathise and Define

Module 2 (09)

216U01E701
Software Design Thiniking
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Module 2: Empathise and Define

- Understanding user needs, pain points, and requirements
- Conducting user research: interviews, surveys, observation
- Creating empathy maps and personas
- Problem framing and defining point-of-view (POV) statements
- Translating requirements into design goals

Empathise and Define

- **User Needs (Empathize & Define)**

Understanding what drives, motivates, and fulfils the user.

Conducting In-Depth Interviews:

- Asking open-ended questions during the Empathize stage to uncover core motivations rather than surface-level opinions.

Observing Natural Behavior:

- Shadowing users in their natural environment to spot implicit needs they may not consciously realize or express verbally.

Empathise and Define

- **Uncovering Emotional Drivers:**
  - Looking beyond functional tasks to understand emotional aspirations (e.g., feeling secure, valued, or efficient).
- **Distinguishing Needs from Wants:**
  - Filtering out superficial feature requests during observation to pinpoint fundamental underlying needs.
- **Mapping Experience Journeys:**
  - Tracking user emotions across touchpoints to see where needs arise, peak, or go unfulfilled.

Empathise and Define

- **Utilizing Empathy Maps:**
  - Synthesizing qualitative research into *Says, Thinks, Does, and Feels* to build holistic user understanding.
- **Identifying Unstated Needs:**
  - Spotting workaround behaviors or hacks users invent to address unmet needs independently.
- **Engaging Extreme Users:**
  - Interviewing both power users and complete novices to reveal edge-case needs that average users adapt to.

Empathise and Define

- **Synthesizing Raw Research Data:**
  - Grouping qualitative notes into key themes during affinity mapping to highlight core human needs.
- **Developing User Personas:**
  - Structuring empathetic insights into representative archetypes that keep team focus centered on authentic needs.
- **Evaluating Contextual Environments:**
  - Analyzing physical, social, and cultural contexts that influence user expectations and behavior.

Empathise and Define

- **Focusing on Latent Desires:**
  - Uncovering needs that users cannot articulate because a solution has never existed before.
- **Identifying Latent Friction:**
  - Noticing micro-frustrations users accept as "normal" during task execution.
- **Translating Observations into Insights:**
  - Converting raw observation data into actionable insights during the Define phase.
- **Framing "How Might We" (HMW) Questions:**
  - Reinterpreting core user needs into HMW prompts that set up empathetic problem formulation.

Empathise and Define

Example of Empathise and Define stage:

- **Agriculture**
  - **Real Users: Farmers, Wholesalers**
- **Weather**
  - **Real Users: Sailors, Farmers, common man**
- **Healthcare**
  - **Real Users: Doctors, Hospital Management Staff, Patients ….**

Empathise and Define

Pain points:

Identifying the specific frustrations, obstacles, and friction users experience.

- **Spotting Friction Points:**
  - Documenting moments where users get stuck, confused, or delayed during their journey.
- **Identifying Financial Pain Points:**
  - Uncovering areas where existing solutions cost users too much money, time, or resources.
- **Uncovering Product/Process Bottlenecks:**
  - Finding structural flaws in current workflows that cause delays or user errors.

Empathise and Define

Pain points:.

- **Mapping Emotional Stress Points:**
  - Tracking moments where users express anxiety, doubt, overload, or fatigue.
- **Detecting Accessibility Barriers:**
  - Identifying physical, technical, or cognitive obstacles that prevent inclusive participation.
- **Analyzing Negative Feedback:**
  - Mining customer support logs, bad reviews, and survey complaints for systemic pain points.

Empathise and Define

Pain points:.

- **Highlighting Disconnected Workarounds:**

- **Categorizing Pain Point Types:**
  - Classifying issues into functional, support, financial, or process-related categories.
- **Determining Severity and Frequency:**
  - Prioritizing pain points based on how frequently they happen and how severe their impact is.

Empathise and Define

Pain points:

- **Assessing Trust and Safety Risks:** Uncovering areas where users feel vulnerable, skeptical, or unsafe using a process.
- **Highlighting Information Gaps:** Spotting moments where users lack critical information to make informed decisions.
- **Setting Boundaries for Problem Definition:** Filtering out irrelevant complaints to focus purely on high-impact pain points.

Empathise and Define

Pain points:

- **Refining Point of View (POV) Statements:** Combining user context and pain points into a clear, concise problem statement.
- **Evaluating Cognitive Load:** Identifying complex interfaces or confusing workflows that cause mental fatigue.
- **Pinpointing Root Causes:** Using root-cause analysis (e.g., 5 Whys) to separate symptoms from true underlying problems.

Empathise and Define

Pain points:

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Empathise and Define

Requirements:

Translating empathetic insights and defined problems into clear, actionable criteria for solutions.

- **Translating Insights into Direct Criteria:** Turning qualitative findings into concrete product, service, or system parameters.
- **Establishing Functional Requirements:** Defining specific actions or capabilities the final solution must accomplish for the user.
- **Setting Non-Functional Criteria:** Specifying performance, usability, accessibility, and security standards based on user context.

Empathise and Define

Requirements:

Translating empathetic insights and defined problems into clear, actionable criteria for solutions.

- **Prioritizing Must-Haves vs. Nice-to-Haves:** Distinguishing critical core requirements from secondary enhancements.
- **Drafting User Stories:** Structuring requirements using human-centered formats (e.g., *"As a [user], I want to [action] so that [benefit]"*).
- **Setting Measurable Success Metrics:** Defining clear benchmarks (e.g., task completion time, error reduction) to measure success.

Empathise and Define

Requirements:

Translating empathetic insights and defined problems into clear, actionable criteria for solutions.

- **Establishing Constraints and Guardrails:** Defining technical, budgetary, or regulatory limits discovered during the empathy phase.
- **Defining Contextual Requirements:** Tailoring requirements to specific environments (e.g., offline functionality, hands-free operation).
- **Ensuring Accessibility Compliance:** Integrating inclusive design standards (e.g., WCAG guidelines, ergonomic factors) into design criteria.

Empathise and Define

Requirements:

Translating empathetic insights and defined problems into clear, actionable criteria for solutions.

- **Creating Design Principles:** Formulating core tenets (e.g., "Simplicity over feature density") to guide technical decisions.
- **Scoping the Minimum Viable Solution (MVS):** Defining the essential feature set required to test and validate ideas in the Next Stage (Ideate/Prototype).
- **Aligning Stakeholder Expectations:** Balancing empathy-driven user requirements with business viability and technical feasibility.

Empathise and Define

Requirements:

Translating empathetic insights and defined problems into clear, actionable criteria for solutions.

- **Mapping Requirement Tracing:** Ensuring every proposed requirement traces directly back to an identified user need or pain point.
- **Standardizing Problem Scope:** Setting strict boundaries to prevent scope creep during the transition to Ideation.
- **Validating Criteria with Users:** Re-confirming defined requirements against initial user feedback to maintain empathetic alignment.

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Conducting User Research

- Conducting user research:
  - Interviews
  - Surveys
  - Observation

Conducting User Research: Interviews

- **Direct interaction with users** :Interviews are qualitative, 1-on-1 conversations designed to understand and to surface personal stories, emotional drivers, and root motivation users' experiences, needs, and challenges.
- **Ask open-ended questions** – Questions such as "What difficulties do you face?" encourage detailed responses rather than simple yes/no answers.
- **Build empathy** – Listening carefully helps designers understand users' emotions, motivations, and expectations.
- **Record important insights** – Notes or recordings help capture key information for later analysis.
- **Support the Define stage** – Information gathered from interviews helps identify the actual problem statement based on user needs.

Conducting User Research: Interviews

1. Focus on Stories over Opinions

- Ask users to describe past experiences rather than hypothetical scenarios (e.g., *"Tell me about the last time you struggled to..."* instead of *"Would you use a tool that..."*). Past behavior reveals real frustration points, whereas future predictions are often overly optimistic.

2. Ask "Why?" to Uncover Hidden Motivations

- Use the "5 Whys" technique to drill past surface-level answers. When a user shares a preference, repeatedly asking *why* uncovers the emotional core—shifting the focus from what they do to how they feel.

Conducting User Research: Interviews

3. Map Verbatim Quotes to Synthesize Pain Points

- Capture exact user quotes during the Empathize stage. Transitioning to the Define stage, cluster these quotes onto an **Empathy Map** (Says, Thinks, Does, Feels) to identify overarching patterns and emotional friction points.

4. Embrace Silence and Non-Verbal Cues

- Pausing after a user answers gives them space to elaborate on deeper thoughts. Pay attention to body language, Hesitations, and tone shifts, as these often point to unspoken anxieties or workaround behaviors.

Conducting User Research: Interviews

5. Reframe Findings into "How Might We" (HMW) Questions

- Use interview insights to transition from raw research to problem framing. Reframe recurring user pain points into actionable HMW statements (e.g., *"HMW make tracking daily routines feel effortless?"*) to set up the Ideation stage.

Conducting User Research: Surveys

- Surveys gather broader quantitative data to validate qualitative findings, measure prevalence, and spot trends across larger user groups.
- **Will help in**
  - **Collecting data from many users**
  - **Using structured questionnaires** – Questions can be multiple-choice, rating scales, or short-answer formats.
  - **Identifying common patterns** – Survey results reveal trends, preferences, and frequently faced problems.
  - **Easy to analyze** – Responses can be summarized using percentages, charts, or statistics.
  - **Help define user needs** – The collected data supports creating a clear and evidence-based problem definition.

Conducting User Research: Surveys

1 Balance Closed and Open-Ended Questions

- Use closed-ended questions (multiple choice, rating scales) to collect quantifiable metrics, and follow up with optional open-ended text fields (e.g., *"Why?"*) to collect qualitative context without fatiguing respondents.

2. Target Specific User Segments

- Screen respondents to ensure you are surveying people who actually match your primary persona or user archetype. Gathering data from the wrong demographic distorts research and leads to misaligned problem definitions.

Conducting User Research: Surveys

3. Keep Surveys Short and Frictionless

- Respect the user's time by limiting surveys to 5–10 clear questions. High completion rates yield cleaner data, while overly long surveys cause rush answers and low-quality responses.

4. Cross-Reference Quantitative Trends with Qualitative Insights

- In the Define stage, use survey statistics (e.g., *"X% of respondents experience ###"*) to back up individual interview stories, proving whether a reported issue is an isolated event or a widespread problem.

Conducting User Research: Surveys

5. Define Problem Scope Using Percentages and Priority Metrics

- Survey data helps narrow down broad problem spaces. Ranking choices reveal what users value most, allowing teams to construct a tight, well-scoped **Problem Statement** focused on high-impact needs.

Conducting User Research: Observations

Study real user behaviour to uncover hidden problems.

Involve watching users in their natural environment to identify unarticulated needs, workarounds, and gaps between what people *say* they do and what they *actually* do.

Conducting User Research: Observations

- **Watch users in real situations** – Observe how users perform tasks in their natural environment.
- **Identify hidden problems** – Users may not mention every issue in interviews, but observation reveals actual behavior.
- **Focus on actions and interactions** – Note how users interact with products, services, or systems.
- **Avoid assumptions** – Observing real behavior provides accurate insights instead of relying only on what users say.
- **Improve problem definition** – Observation helps validate findings and ensures the Define stage focuses on real user challenges.

Conducting User Research: Observations

1. Adopt a "Beginner's Mindset"

- Enter the environment without pre-existing assumptions or biases. Watch users interact with their environment as if seeing it for the first time, noting friction points that users themselves might take for granted.

2. Document Workarounds and Hacks

- Pay close attention to improvised user behaviors (e.g., how user remembers the things or create shortcuts etc). These workarounds highlight existing system failures and point directly to design opportunities.

Conducting User Research: Observations

3. Capture the Context (Physical and Digital)

- Take detailed notes on environmental factors like distractions, lighting, tool layouts, or network speeds. Understanding the context prevents designing solutions that only work in idealized conditions.

4. Use Structured Observation Frameworks (e.g., basics of users behaviours)

- Systematically organize findings across five categories: **A**ctivities, **E**nvironments, **I**nteractions, **O**bjects, and **U**sers. This prevents biased note-taking and ensures balanced data collection.

Conducting User Research: Observations

5. Synthesize Observations to Identify Core Contradictions

- In the Define stage, compare observed behavior against interview statements. Discrepancies (e.g., a user claiming a process is easy, yet struggling for 10 minutes) reveal hidden needs that form the foundation of compelling **Point of View (POV)** statements.

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Creating empathy maps and personas

- **4 quadrants**
- Open ended
- Bottom-up approach
- Useful in creating personas

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Problem framing & POVs

- A poorly framed problem usually leads to building the wrong product or service—even if the design itself is executed flawlessly

- **Stops wrong solutions:** Ensures teams do not fix surface symptoms while missing the real issue.
- **Saves time and money:** Limits scope creep and prevents costly rework on misdirected ideas.
- **Aligns the team:** Creates a shared vision and clear goal for everyone involved.
- **Drives true innovation:** Turns vague frustrations into actionable, human-centred opportunities.

Problem framing

Steps in problem framing

1: Gather User Insights

2: Map the System

3: Question the Initial Brief

4: Write a Point of View (POV) Statement

5: Launch Generative Questions

Problem framing

- **Step 1: Gather User Insights**
  - **Conduct interviews:** Speak with the people facing the issue.
  - **Observe actions:** Watch how users interact with current workarounds.
  - **Map empathy:** Document what users say, do, think, and feel.
- **Step 2: Map the System**
  - **Trace journeys:** Chart every touchpoint a user encounters.
  - **Identify bottlenecks:** Pinpoint exactly where frustration occurs.
  - **Layer context:** Look at cultural, technical, and financial factors.
- **Step 3: Question the Initial Brief**
  - **Ask "Why?":** Use the 5 Whys technique to dig past symptoms.
  - **Find root causes:** Separate the immediate complaint from the deep trigger.
  - **Check assumptions:** Verify if the "known" facts are backed by data.

Problem framing

- **Step 4: Write a Point of View (POV) Statement**
  - **Pick a persona:** Define the specific user group.
  - **State the need:** Describe what they deeply require.
  - **Share the insight:** Explain the unexpected truth you discovered.
  - **Use this template:** "[User] needs [Need] because [Insight]."
- **Step 5: Launch Generative Questions**
  - **Flip the POV:** Turn your statement into "How Might We" (HMW) questions.
  - **Check the scope:** Ensure questions are narrow enough to focus, but broad enough for creativity.
  - **Select the best:** Choose 2–3 HMW questions to kickstart your brainstorming.

POVs

- A POV statement focuses on a specific user's needs and challenges.
- Clearly explains what the user requires or wants to achieve.
- Highlights the important observations that explain why the user has that need.
- Gives the team a clear focus for generating ideas.
- Precise and easy to use during ideation.
- Used to create brainstorming questions for innovation in the form of *How Might We* solve the problem.

POVs

Creation of POV using User + Need + Insight

- **User:**
  - Identifies the specific person or group for whom the design solution is intended.
  - Not a generic audience but a clearly defined persona based on empathy research.
- **Need:**
  - Represents what the user is trying to accomplish or the challenge they face.
  - Expressed as verbs rather than nouns, emphasizing action and purpose.
- **Insight**
  - Explains why the need exists and provides a deeper understanding of the user's behaviour, emotions, or constraints.
  - Derived from empathy work — interviews, observations, and user research — and often reveal hidden pain points or opportunities.

POVs

Case study:

A solution is needed for booking hotels online.

Personas identified:

- Business Travellers
- Budget Travellers
- Professional planner agents
- Hotel owners

Type of User	User	Need	Insight
Business Travellers	A corporate business traveller managing tight schedules and dynamic itineraries.	To rapidly reserve rooms near specific work venues with automated expense-ready receipts and flexible cancellation policies.	Uncontrollable meeting delays make booking speed and friction-free changes far more critical to them than finding the cheapest rate or luxury amenities.
POV Statement: A frequent business traveller who arrives in a new city late at night needs a 1-click booking process with guaranteed instant confirmation, because flight delays leave them exhausted and anxious about finding a safe room immediately.			

Type of User	User	Need	Insight
Budget Travellers	A budget traveller looking for cost effective price of the hotel	Know the total cost , with amenities	Many a times the hidden costs are not included, so difficult to get final cost
POV Statement: A budget-conscious solo traveller needs a clear, transparent way to see the *total final price* of a hotel room upfront, because hidden fees and unexpected checkout charges make them feel deceived and destroy trust in the platform.			

Type of User	User	Need	Insight
Professional planner agents	Agents who assist in bookings	Know different options and availability and commission	Provide better hotels in reasonable rates and still getting the appropriate commission
POV Statement: Professional agents assisting people to book hotels in reasonable price and still getting decent commission.			

POV to How We Might Questions

Once problem framing and POVs are identified convert them into actional design

HWM :

Business Travellers:

- streamline the booking process into a single-click experience for frequent travellers on tight schedules
- simplify receipt generation and expense-reporting integration so business travellers can submit claims effortlessly
- guarantee last-minute availability and flexible cancellation policies for travellers dealing with unpredictable flight or meeting schedules

POV to How We Might Questions

Once problem framing and POVs are identified convert them into actional design

HWM :

Budget Travellers:

- **D**isplay all hidden fees, taxes, and final costs upfront so budget travellers never experience checkout price shock
- **H**elp cost-conscious travellers easily discover, compare, and lock in the best value deals across flexible date ranges
- **H**ighlight low-cost amenities (free Wi-Fi, breakfast, nearby public transit) so users can accurately evaluate total trip expenses

POV to How We Might Questions

Once problem framing and POVs are identified convert them into actional design

HWM :

Professional Agents:

- **B**uild a dedicated dashboard for agents to manage, track, and modify multiple room bookings and bulk reservations seamlessly
- **P**rovide agents with instant client-shareable quotes and customized itineraries without requiring repetitive manual data entry
- **A**utomate commission tracking and confirmation status updates to save travel agents time and reduce administrative back-and-forth

POV to How We Might Questions

Once problem framing and POVs are identified convert them into actional design

HWM :

Hotel Owners:

- **P**rovide hotel owners with intuitive tools to manage live room inventory and dynamically update pricing based on local demand
- **H**elp them to showcase their unique property highlights and verified guest reviews to stand out against larger chains
- **M**inimize drop-offs during checkout and reduce last-minute room cancellations for hotel managers

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Translating Requirements into design goals

Raw, unstructured information—such as user feedback, business targets, technical constraints, and empathy research is converted into clear, actionable objectives that direct the design process.

Translating Requirements into design goals

1. **Gather Requirements**
    1. Collect user needs, stakeholder expectations, and business requirements through interviews, surveys, observations, and research.
2. **Analyze the Requirements**
    1. Study the collected information to understand user problems, expectations, and project constraints.
3. **Identify User Needs**
    1. Separate the essential needs from optional features and focus on solving the users' main problems.
4. **Prioritize Requirements**
    1. Rank the requirements based on importance, user impact, feasibility, cost, and project objectives.

Translating Requirements into design goals

5. **Define the Problem Clearly**
    1. Write a clear problem statement that describes what needs to be solved before creating design goals.
6. **Convert Needs into Goals**
    1. Translate each requirement into a specific design objective.
    2. **Example:**
        1. Requirement: "Users find the app difficult to navigate."
        2. Design Goal: "Create a simple and intuitive navigation system."
7. **Make Goals SMART**
    1. Ensure each design goal is:
        1. **S**pecific
        2. **M**easurable
        3. **A**chievable
        4. **R**elevant
        5. **T**ime-bound

Translating Requirements into design goals

8. **Align Goals with Business Objectives**
    - Ensure the design goals support organizational goals such as improving customer satisfaction, increasing efficiency, or reducing costs.
9. **Validate the Design Goals**
    - Review the goals with users, stakeholders, and the design team to confirm they address the identified requirements.
10. **Review and Refine**
    - Update the design goals based on user feedback, prototype testing, and changing project requirements to keep them effective.

Translating Requirements into design goals

Persona	Requirements	Design Goals
Budget Travellers	• Affordable hotel options • Price comparison and discounts • Filters for budget, ratings, and facilities • Genuine customer reviews • Easy cancellation and refund options	• Make it easy to compare hotel prices. • Display offers and discounts prominently. • Provide transparent pricing with no hidden charges. • Help users make informed decisions through reviews and ratings. • Simplify cancellation and refund processes..

Translating Requirements into design goals

Persona	Requirements	Design Goals
Business Travellers	• Fast hotel search and booking • Hotels near business locations • Flexible check-in/check-out options • Business-friendly amenities (Wi-Fi, meeting rooms) • Secure payment and instant invoice	• Minimize booking time with a simple workflow. • Provide location-based hotel recommendations. • Highlight business amenities clearly. • Enable secure, quick payments and automatic invoice generation. • Support flexible booking modifications.

Translating Requirements into design goals

Persona	Requirements	Design Goals
Professional Planners	• Manage bookings for multiple clients • Bulk booking capability • Client profile management • Booking history and reports • Easy communication with hotels	• Design an efficient dashboard for managing multiple bookings. • Enable quick client and reservation management. • Provide detailed booking reports and tracking. • Support bulk booking with minimal effort. • Improve communication between agents and hotels.

Translating Requirements into design goals

Persona	Requirements	Design Goals
Hotel Owners	• Add and update hotel details • Manage room availability and pricing • Receive booking notifications • View customer reviews and ratings • Access sales and occupancy reports	• Create an easy-to-use hotel management dashboard. • Allow real-time updates of rooms and prices. • Provide instant booking notifications. • Display analytics for occupancy, revenue, and customer feedback. • Enable efficient management of hotel operations.